FILTER DESIGN TEST CASE

Low-Pass Filter - WiFi 2.4 GHz for 5G NR Sub-6 GHz Front-End Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Low-Pass Filter filter targeting WiFi 2.4 GHz for 5G NR Sub-6 GHz Front-End Module. The filter is designed on AlN on Si substrate with a target center frequency of 4423.4 MHz and bandwidth of 137.7 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 5G NR Sub-6 GHz Front-End Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Low-Pass Filter
- Frequency Band: WiFi 2.4 GHz
- Application: 5G NR Sub-6 GHz Front-End Module
- Substrate: AlN on Si
- Package: CSP (Chip Scale Package)
- Design Tool: PathWave EM Design
- Design Center: Helsinki 5G Lab

This specification applies to all variants of the Low-Pass Filter design targeting WiFi 2.4 GHz. Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 4423.4 MHz
Bandwidth (3 dB): 137.7 MHz
Insertion Loss Target: <= 2.4 dB
Return Loss Target: >= 14.0 dB
Out-of-Band Rejection: >= 36.0 dB
Isolation (Duplexer): >= 45.0 dB
Q Factor: 4098
Temperature Coefficient: -27.7 ppm/°C
Die Size: 1.3 x 1.9 mm
Wafer Size: 6-inch
Number of Resonators: 4
Electrode Material: Pt/Al

Passivation: SiO2

4. TEST PROCEDURE

1. Open PathWave EM Design and load the Low-Pass Filter template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 4423.4 MHz, BW 137.7 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.4 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, AlN on Si).
10. Perform on-wafer probing using Rohde & Schwarz ZNB40.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 13270 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.4 dB
- Return loss in passband shall be >= 14.0 dB
- Out-of-band rejection at +/- 207 MHz offset >= 36.0 dB
- Group delay variation across passband shall be <= 10 ns
- Temperature drift over -40°C to +85°C shall be <= 15.3 MHz
- Power handling shall be >= +26 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 89%

6. TEST RESULTS

Measurement Date: 2025-06-03
Devices Tested: 51
Insertion Loss (meas): 2.14 dB
Return Loss (meas): 16.0 dB
Rejection (meas): 39.3 dB
Isolation (meas): 50.0 dB
Q Factor (meas): 4098
Group Delay Variation: 3.9 ns
Power Handling: +28 dBm
Die Yield: 84.0%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

a) The Low-Pass Filter design demonstrated excellent performance for WiFi 2.4 GHz.
b) Insertion loss met target specification.
c) Out-of-band rejection exceeded the minimum requirement.
d) Package parasitic effects were consistent with 3D EM model predictions.
e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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Reviewer:	Dr. Pierre Dubois	Date: 2025-06-08
Approver:	Dr. Karen Mitchell	Date: 2025-06-07

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